Econ 812 Recitation 3 Notes

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1 Value Functions

1.1 Bellman Equation

The Bellman Equation is as follows:

$$V_t(s_t) = \max_{a_t}\{u(s_t, a_t) + \beta V_{t+1}(s_{t+1})\} \tag{1}$$

$$s.t. : a_t \in \Gamma(s_t) \tag{2}$$

$$s_{t+1} = \psi(s_t, a_t, z_t, z_{t+1}) \tag{3}$$

Where the action at time t is in by some set that depends on the space (say, you can't consume 0 or less than your savings if there are borrowing constraints). And the transition equation could depend on the state, action, or exogenous components

1.2 A simple example

Imagine a world with three states (Figure 1) and two periods. Say, we are dropped into this world in the middle at U=2 at T=1 and can only move (freely) once before T=2.

What we are trying to accomplish is to maximize the value of our states. But we need some guiding light on how to make the optimal choice of our action (moving once).

This is where the policy function comes in. It is the policy that our actions follow to give the optimal answer. Now, there could be many policy functions. But we set up the problem to have a unique solution.

Some examples of policy functions might be "always left" or "always right" or "don't move" or "move right half the time, move left half the time"

$$E[\text{“left”}] = E[U(\text{now})] + E[U(\text{later})] = 2 + 1 = 3 \tag{4}$$

$$E[\text{“right”}] = E[U(\text{now})] + E[U(\text{later})] = 2 + 3 = 5 \tag{5}$$

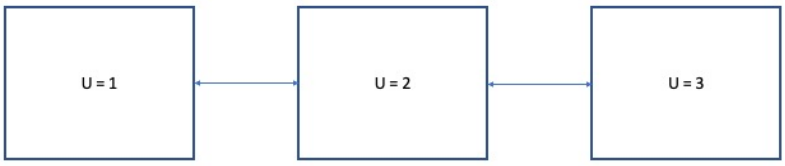


Figure 1: Three states of the world with utility equalling 1, 2, and 3

$$E[\text{“1/2 left} + 1/2 \text{ right”}] = 2 + 1/2 + 3/2 = 4 \tag{6}$$

We pick here the policy that gives the highest expected value, and say that this is the value of getting dropped in this world in the middle. We have uncovered the policy function and determined the (maximum) value of the state.

Now let's apply this idea to DSGE models, with a numerical example.

1.3 General Steps to DP model

When solving these types of problems, you should follow these steps:

Step 1 Substitute constraint for c

Step 2 Solve for policy function

Step 3 Plug policy function back into Bellman

Step 4 Match up coefficients

1.4 Finite Horizon DP Example

Solving the same problem as before:

$$\max_{c_t, k_{t+1}} \sum_{t=0}^{T} \beta^t u(c_t)$$

Subject to

$$c_t + k_{t+1} \leq z_t k_t^{\alpha}$$

Letting $T = 3$, $\beta = 0.8$, $u(C) = \log C$, $z_t = 1$ and $\alpha = 0.5$.

We know in the last period that $K_{T+1} = K_4 = 0$ so $C_T = K_T \rightarrow C_3 = K_3^\alpha$
Returning to the Bellman Equation...

$$V_3(K_3) = u(C_3) + \beta V_4(K_4) = \log C_3 \tag{7}$$

With the constraints of

$$C_3 + K_4 = K_3^\alpha \tag{8}$$

Plugging in we have

$$V_3(K_3) = \log K_3^\alpha = \alpha \log K_3 \tag{9}$$

Guess and Verify that $V_T = A_T + B_T \log K_T$. So we have that $A_3 = 0$ and $B_3 = 0.5$

Writing out the value function for period 2 (including Step 1 plug in constraint for C_2)

$$V_2(K_2) = \log(K_2^\alpha - K_3) + \beta[A_3 + B_3 \log K_3] \tag{10}$$

Then solving for policy function (Step 2)

$$\frac{-1}{K_2^\alpha - K_3} + \beta B_3 \frac{1}{K_3} = 0 \tag{11}$$

$$\beta B_3 K_2^\alpha = K_3 + \beta B_2 K_3 \tag{12}$$

$$K_3 = \frac{\beta B_3 K_2^\alpha}{1 + \beta B_3} \tag{13}$$

Policy function for $T = 2$

Plugging back into the value function (Step 3)

$$V_2(K_2) = \log(K_2^\alpha - \frac{\beta B_3 K_2^\alpha}{1 + \beta B_3}) + \beta[A_3 + B_3 \log(\frac{\beta B_3 K_2^\alpha}{1 + \beta B_3})] \tag{14}$$

Likewise for V_1 and V_0

$$V_1(K_1) = \log(K_1^\alpha - \frac{\beta B_2 K_1^\alpha}{1 + \beta B_2}) + \beta[A_2 + B_2 \log(\frac{\beta B_2 K_1^\alpha}{1 + \beta B_2})] \tag{15}$$

$$V_0(K_0) = \log(K_0^\alpha - \frac{\beta B_1 K_0^\alpha}{1 + \beta B_1}) + \beta[A_1 + B_1 \log(\frac{\beta B_1 K_0^\alpha}{1 + \beta B_1})] \tag{16}$$

Expand out terms and match up coefficients (Step 4)

$$V_2(K_2) = A_2 + B_2 \log K_2 = \underbrace{\log(1 - \frac{\beta B_3}{1+\beta B_3}) + \beta A_3 + \beta B_3 \log(\frac{\beta B_3}{1+\beta B_3})}_{A_2} + \underbrace{\alpha}_{B_2} \log K_2 + \underbrace{\beta\alpha B_3}_{B_2} \log K_2 \tag{17}$$

Plugging in $\beta = 0.8$ $A_3 = 0$ and $B_3 = \alpha = 0.5$ to find $A_2 = -0.84$ and $B_2 = 0.7$. Continue on for V_1

$$V_1(K_1) = A_1 + B_1 \log K_1 = \underbrace{\log(1 - \frac{\beta B_2}{1+\beta B_2}) + \beta A_2 + \beta B_2 \log(\frac{\beta B_2}{1+\beta B_2})}_{A_1} + \underbrace{\alpha}_{B_1} \log K_1 + \underbrace{\beta\alpha B_2}_{B_1} \log K_1 \tag{18}$$

$$\underbrace{\log(1 - \frac{0.8*0.7}{1+0.8*0.7}) + 0.8 * -0.84 + 0.8*0.7 \log(\frac{0.8*0.7}{1+0.8*0.7})}_{A_1} + \underbrace{0.5}_{B_1} \log K_1 + \underbrace{0.8*0.5*0.7}_{B_1} \log K_1 \tag{19}$$

$A_1 = -1.69$

$B_1 = 0.78$

$A_0 = -2.43$

$B_0 = 0.812$

Now we have recovered all of the value function coefficients. To recover the state and choice variables, start with an initial state $K_0 = 10$ and then use the policy function (36) to calculate through time.

$$K_1 = \frac{0.8 * 0.78 * 10^{1/2}}{1 + 0.8 * 0.78} = 1.22 \tag{20}$$

$K_2 = 0.4$

$K_3 = 0.18$

$K_4 = 0$

Can use the resource constraint to find the values of consumption as well.